

SIH26 MINE SUBSIDENCE MONITORING & PREDICTION PLATFORM

Master/Slave Node Architecture, Speech-Ready Hinglish Presentation Pitch & Technical Guide

1. Master Node vs Slave Node Mesh Architecture

Our hardware topology uses a **Master-Slave Wireless Sensor Mesh** designed specifically for hostile underground mining environments where internet access is unavailable on the surface panels.

Component Layer	Slave Nodes (Field Sensor Nodes N1 - N5)	Master Node (Central Gateway Node)
Hardware Unit	ESP32-S3 Microcontroller + LiFePO4 Solar Battery	Industrial Raspberry Pi 4 / ESP32 Gateway + SIM7000 4G LTE
Sensors Attached	IMU (Tilt), Strain Gauge, Geophone (Vib), Crack Wire, BME280	Local Relay Siren, Flash Memory Storage, SIM7000 LTE Cellular Module
Primary Role	Collects raw sensor streams at 100Hz, extracts 8 features, runs Stage 1 Edge TinyML (Isolation Forest) in microsecond C header.	Aggregates telemetry from all 5 Slave Nodes, runs Stage 2 Gateway Spatial Correlator , triggers offline alarms, uplinks to Cloud.
Communication	Transmits compact 8-feature packets over ESP-NOW / 802.15.4 Radio Mesh to Master Node.	Receives mesh radio packets from Slaves; uplinks 24-hour sequence batches to Stage 3 Cloud LSTM server via LTE.
Fault Tolerance	If radio link to Master breaks, Slave operates standalone, triggering local piezobuzzer on anomaly.	Stores up to 30 days of offline telemetry in local SQLite DB if cellular tower goes down.

2. Speech-Ready Hinglish Presentation Pitch (SIH Jury Demo)

[Slide 1: Intro & Problem Statement]

"Respected Judges and Team, Good Morning! Aaj hum solve kar rahe hain mining industry ki sabse dangerous problem: **Underground Mine Subsidence and Sudden Ground Collapse**.

Jab underground coal ya metal mines mein roof collapse hota hai, toh surface par bina kisi warning ke zameen dhas jaati hai. Iss se heavy equipment destruction, roads damage, aur human lives ki loss hoti hai.

Traditional systems (jaise Satellite InSAR) 12-day late hote hain aur manual surveys real-time warnings nahi de pate. Humne banaya hai **AI-Enabled Low-Cost Real-Time Mine Subsidence Monitoring Platform — a 3-Stage Hierarchical Master-Slave System!**"

[Slide 2: Master/Slave & 3-Stage Architecture]

"Humare system ke 3 layers hain:

1. Slave Nodes (Stage 1 Edge TinyML): Field mein har sensor point par **ESP32 Slave Node** laga hai jisme Isolation Forest Anomaly Model C-header mein quantize hoke chalta hai. Ye microsecond execution karta hai local node par.

2. Master Node (Stage 2 Gateway Correlator): Underground mines mein explosive blasting hoti hai. Master Node saare Slave Nodes (N1 to N5) ka data gather karke **Blast vs True Subsidence** ko discriminate karta hai taaki false siren na baje.

3. Cloud Server (Stage 3 Multi-Output LSTM): Cloud par Multi-Head Deep LSTM Network chalta hai jo 24-hour sequence process karke **Subsidence Risk %, Max Displacement (in mm), aur Severity Class (Low/Med/High)** predict karta hai."

[Slide 3: Dataset & Physics Simulator]

"Judges, real mine collapse ka public dataset available nahi hota. Isliye humne ek **Physics-Based Geomechanical Sensor Simulator** build kiya jo 90-days ka continuous telemetry stream generate karta hai across 5 Slave Nodes (N1 to N5). Data mein 4 phases hain: Normal, Pre-Subsidence Creep (0.1 to 0.5 deg tilt), Active Subsidence (>1.5 deg tilt + broken crack wire), and Operational Blasting. Iss dataset par humne saare models benchmark kiye hain!"

3. Detailed Visualizations & Performance Graphs Breakdown

Graph 1: Cloud Model LSTM Evaluation (cloud_model_evaluation.png)

Plot Title	Visual Appearance	Technical Meaning & Explanation
Displacement Estimation (MAE)	Blue scatter dots around red 1:1 diagonal line	Actual vs Predicted Displacement (mm). Red line ideal prediction ko darshata hai. Points red line ke paas hain with MAE of ~2.4 mm, proving sub-centimeter displacement accuracy.
Severity Confusion Matrix	3x3 Heatmap grid (Low, Medium, High)	Model classification accuracy. Dark blue diagonal confirms High-risk active subsidence events zero times Low-risk class mein misclassify huye hain.
Cloud Risk Trajectory Over Time	Red risk curve crossing 0.5 warning line	90-Day Timeline Prediction. Normal phase mein risk < 0.1 rehta hai. Day 60-75 (Pre-subsidence) mein risk rise hokar 0.5 threshold cross karta hai, giving hours of early warning.
Displacement Residual Error	Green bell curve centered at 0.0 mm	Prediction error distribution. Gaussian curve zero par centered hai, proving model is completely unbiased.

Graph 2: Edge Isolation Forest Evaluation (edge_model_evaluation.png)

Plot Title	Visual Appearance	Technical Meaning & Explanation
Score Distribution	Green (Normal) vs Red (Anomaly) split by -0.61 line	Anomaly score separation. Normal sensor readings right side par drop hote hain, structural anomalies left side par. Clear margin proves strong edge accuracy.
ROC Curve	Blue line hugging top-left corner (AUC > 0.99)	Trade-off between True Positive Rate and FPR. High AUC (>0.99) proves edge model generates negligible false alarms.
Scores Over Time (Node N3)	Scatter plot: Green -> Orange -> Red points	Center Slave Node N3 anomaly scores over 90 days. Day 60 par pre-subsidence creep shuru hote hi scores red threshold line ke neeche drop ho jate hain.

Graph 3: 90-Day Sensor Mesh Telemetry (dataset_sensor_streams.png)

Subplot Title	Visual Feature	Technical Meaning & Explanation
Ground Tilt Deformation	Red line (N3 Center) vs Blue line (N1 Edge)	Spatial deformation profile. Center node N3 shows tilt rising from 0.1 deg to >1.5 deg, while edge node N1 stays flat.
Strain Gauge Stress Delta	Orange curve rising during pre-subsidence	Tensile strain (microstrain) buildup before rock fracture occurs.
Vibration & Blasting	Green line with sharp vertical spikes	Operational blasts show transient acceleration spikes without progressive tilt growth.
Crack Breakwire State	Step signal dropping from 1 to 0	Physical tripwire circuit. Jab surface rock split hoti hai, wire break hoti hai and status 0 ho jata hai.

4. Backend REST API Endpoints & Request Payloads Line-by-Line

1■■ Edge Prediction Endpoint: POST /api/predict/edge

```
{
  "tilt_mean": 0.02,           // Ground baseline tilt angle (degrees). Normal stable ground < 0.1 deg.
  "tilt_rate": 0.0001,        // Angular deformation speed (deg/min). Spikes indicate active movement.
  "strain_delta": 0.5,         // Strain gauge stress measurement (microstrain). Tensile rock stretch.
  "vib_rms": 0.025,           // Root-Mean-Square vibration acceleration (g). Ambient seismic background noise.
  "vib_peak": 0.06,           // Maximum peak vibration acceleration (g).
  "vib_dominant_freq": 14.5,   // Dominant frequency (Hz). Subsidence seismic movement is low freq (5-20Hz).
  "crack_status": 1,           // Physical tripwire breakwire. 1 = Intact wire, 0 = Rock split/severed wire.
  "temp_humidity_index": 0.55 // Environmental index (0-1). Rules out thermal expansion false positives.
}
```

2■■ Gateway Spatial Correlation Endpoint: POST /api/predict/gateway

```
{
  "packets": {
    "N1": { "tilt_mean": 0.01, "vib_rms": 0.02, "strain_delta": 0.2 },
    "N2": { "tilt_mean": 0.15, "vib_rms": 0.05, "strain_delta": 3.4 },
  }
}
```

```

    "N3": { "tilt_mean": 0.45, "vib_rms": 0.12, "strain_delta": 8.9 }, // Center node showing max stress
    "N4": { "tilt_mean": 0.12, "vib_rms": 0.04, "strain_delta": 2.8 },
    "N5": { "tilt_mean": 0.01, "vib_rms": 0.02, "strain_delta": 0.1 }
  }
}
// Logic: If ALL nodes (N1..N5) show high vib_rms without tilt creep -> GATEWAY CLASSIFIES AS OPERATIONAL BLAST.
//        If ONLY center nodes (N2..N4) show tilt/strain buildup -> GATEWAY TRIGGERS SUBSIDENCE ALERT.

```

3 Cloud LSTM Forecasting Endpoint: POST /api/predict/cloud

```

{
  "sequence": [
    [0.01, 0.0001, 0.2, 0.02, 0.05, 12.0, 1.0, 0.5, ... 40 spatial features ...],
    ... 288 timesteps for 24-hour rolling sequence window ...
  ]
}
// Response:
{
  "subsidence_probability": 0.8924, // 89.24% risk of ground collapse in 24-48 hours
  "predicted_max_displacement_mm": 42.5, // 42.5 mm estimated surface sinking depth
  "severity_class": "HIGH", // Severity level alert for control room dashboard
  "confidence": 0.9102 // Model confidence rating
}

```

5. Technical Model Rationale & System Execution Guide

Why These Specific ML Models Were Chosen

Stage	Model Architecture	Why Chosen? (Technical Rationale)	How it Solves the Problem
Stage 1 (Edge)	Isolation Forest (C-Header Quantized)	Low RAM/compute footprint. No matrix inversions or neural weights. Easily converted to C header (isolation_forest_model.h) for ESP32 Slave Nodes.	Microsecond execution directly on field Slave Nodes. Zero network dependency to detect immediate rock split & tilt anomalies.
Stage 2 (Gateway)	Spatial Correlator (Grid Rule Engine)	Statistical multi-node spatial correlation across grid geometry (N1 to N5).	Eliminates false alarms caused by mining explosive blasts by verifying multi-node spatial vibration signatures.
Stage 3 (Cloud)	Multi-Output LSTM (Multi-Head Recurrent)	LSTMs capture temporal sequence dependencies over 24-hour rolling windows (288 timesteps).	Multi-head output simultaneously forecasts risk probability %, numeric displacement in mm, and severity level (Low/Med/High).

How to Run & Demonstrate the System

Step 1: Launch Backend Server (using project virtual environment)

Open terminal in project root and execute:
d:\innertask\ml_venv\Scripts\python.exe d:\innertask\mine_subsidence_ml\web_app\server.py

Step 2: Access Control Room Web Dashboard & Swagger UI

- **Web UI Dashboard:** Open <http://localhost:8050/> in your browser.
- **Interactive API Docs (Swagger UI):** Open <http://localhost:8050/docs> to test live API payloads.
- **API Health Endpoint:** Open <http://localhost:8050/api/health> to verify model loading status.